Business Objectives and Research Hypotheses

Business Objective 1: To identify category- and segment-level drivers of first purchase conversion and overall sales growth

Question 1: Which product categories and subcategories have the highest add-to-cart to purchase conversion and lowest cancellation rates?

Hypothesis: Certain product categories and subcategories achieve significantly higher conversion rates and lower cancellation rates than others.

Question 2: How do conversion rates vary by customer segment (New vs. Loyal), gender, and age group?

Hypothesis: Loyal customers and middle-aged customer groups exhibit higher purchase conversion rates compared to new and younger customers.

Question 3: How do sales revenues vary across different months?

Hypothesis: Sales revenue demonstrates seasonal fluctuations, with specific months generating significantly higher revenue than others.

Question 4: How has overall sales performance changed over the years?

Hypothesis: Overall sales revenue and order volume have shown a positive growth trend over time.

Business Objective 2: To understand repeat purchase, volume, and loyalty patterns across demographics, categories, and delivery performance

Question 1: Which categories and brands most strongly predict repeat purchases among New customers transitioning to Loyal?

Hypothesis: Customers purchasing from specific categories and brands are more likely to make repeat purchases and become loyal customers.

Question 2: How do discount rates, shipping costs, and unit prices affect reorder probability within 30/60/90 days?

Hypothesis: Higher discounts and lower shipping costs increase the likelihood of repeat purchases within 90 days.

Question 3: Which Categories Have the Top 10 Highest Order Volume?

Hypothesis: A small number of product categories account for a disproportionately large share of total order volume.

Question 4: What is the Repeat Purchase Rate?

Hypothesis: Repeat purchases contribute a substantial portion of total orders, indicating the presence of a loyal customer base.

Business Objective 3: To optimize product portfolio, pricing, discounting, and shipping for profitable growth

Question 1: What is item-level margin after discount and shipping, by category, brand, and stock availability?

Hypothesis: Profit margins vary significantly across product categories and brands after accounting for discounts and shipping costs.

Question 2: How do changes in discount bands influence conversion and average order value across segments?

Hypothesis: Moderate discount levels improve conversion rates and average order value more effectively than very low or very high discount levels.

Question 3: Which Product Categories Generate the Top 10 Highest Revenue?

Hypothesis: Revenue generation is concentrated within a limited number of high-performing product categories.

Question 4: How does product pricing influence customer satisfaction and review ratings?

Hypothesis: Higher-priced products tend to receive better customer ratings due to perceived quality and value.

Business Objective 4: To leverage reviews, ratings, and customer sentiment to enhance conversion and retention

Question 1: Which categories and brands consistently earn high ratings and low returns/cancellations?

Hypothesis: Products with consistently high ratings experience lower cancellation rates and stronger customer retention.

Question 2: How do review scores correlate with reorder rates and time-to-next-purchase?

Hypothesis: Customers who provide higher review scores are more likely to repurchase and do so within shorter time intervals.

Question 3: Which Product Categories Receive the Best Reviews?

Hypothesis: Certain product categories consistently achieve higher average review scores than others.

Question 4: How do positive vs. negative review trends impact overall category conversion rates?

Hypothesis: Categories with predominantly positive reviews exhibit higher conversion rates than categories with negative review trends.

Business Objective 5: To evaluate logistics and operational efficiency to improve customer experience

Question 1: How Does Delivery Performance Affect Customer Satisfaction?

Hypothesis: Faster and on-time deliveries are associated with higher customer satisfaction ratings.

Question 2: How do shipping costs and delivery times vary across product categories and customer regions?

Hypothesis: Shipping costs and delivery times differ significantly across product categories and geographic regions.

Question 3: What impact do delivery delays have on order cancellation and product return rates?

Hypothesis: Orders experiencing delivery delays have higher cancellation and return rates than on-time deliveries.

Question 4: How does fulfillment speed affect customer review ratings and overall satisfaction scores?

Hypothesis: Shorter fulfillment times lead to higher review ratings and overall customer satisfaction.

Business Objective 6: To optimize customer retention and lifetime value across key product categories

Question 1: What are the primary drivers of customer churn across high-volume product categories?

Hypothesis: Low customer ratings, higher prices, and delivery delays are associated with increased customer churn.

Question 2: How does the time elapsed between initial and second purchases differ across product categories?

Hypothesis: Customers purchasing from frequently consumed product categories make their second purchase sooner than customers in other categories.

Question 3: Which customer segments generate the highest total revenue over their purchasing lifetime?

Hypothesis: Loyal customers contribute significantly higher lifetime revenue than new or occasional customers.

Question 4: How do targeted discounts and loyalty incentives impact re-engagement rates for inactive buyers?

Hypothesis: Customers receiving discounts and promotional incentives are more likely to return and make subsequent purchases.